Paul G. Beckman

Updated October 15, 2026 **CV**

Education

PhD in Mathematics *Courant Institute, New York University* 2020-2025
Advisor: Michael O'Neil
Thesis: Fast transform methods for Gaussian random fields

BS in Computational and Applied Mathematics *University of Chicago* 2015-2019
Advisor: Mihai Anitescu
Thesis: Nonstationary Gaussian process approximations of piecewise analytic computer codes

Research Positions

Peter O'Donnell Jr. Postdoctoral Fellow *Oden Institute, UT Austin* 2025-present
Advisors: Joe Kileel, Per-Gunnar Martinsson

CSGF Practicum *Lawrence Berkeley National Laboratory* 2023
Advisors: Yang Liu, Xiaoye Sherry Li

Mathematics and Statistics Researcher *Argonne National Laboratory* 2019-2020
Advisors: Mihai Anitescu, Michael Stein

Publications and Preprints

Persson, David, **Paul G. Beckman**, Tyler Chen, Diana Halikias, Christopher Musco. "A Recursive Butterfly Factorization with Optimality Guarantees." arXiv preprint arXiv:2607.29361 (2026).

Beckman, Paul G., Samuel F. Potter, Michael O'Neil. "A Butterfly-Accelerated Manifold Harmonic Transform." arXiv preprint arXiv:2605.21828 (2026).

Beckman, Paul G., Michael O'Neil. "A Nonuniform Fast Hankel Transform." *SIAM Journal on Scientific Computing*, no. 2 (2026): A784-A803.

Geoga, Christopher J., **Beckman, Paul G.**. "Fast nonparametric spectral density estimation from irregularly sampled data." *SIAM Journal on Scientific Computing*, To appear.

Beckman, Paul G., Christopher J. Geoga. "Fast Adaptive Fourier Integration for Spectral Densities of Gaussian Processes." *Statistics and Computing* 34, no. 6 (2024): 217.

Beckman, Paul G., Christopher J. Geoga, Michael L. Stein, and Mihai Anitescu. "Scalable Computations for Nonstationary Gaussian Processes." *Statistics and Computing* 33, no. 4 (2023): 84.

Williams–Young, David B., **Paul G. Beckman**, and Chao Yang. "A Shift Selection Strategy for Parallel Shift-Invert Spectrum Slicing in Symmetric Self-Consistent Eigenvalue Computation." *ACM Transactions on Mathematical Software (TOMS)* 46, no. 4 (2020): 1-31.

Awards

Moses A. Greenfield Research Prize *Courant Institute* 2024
Computational Science Graduate Fellowship *Department of Energy* 2020

Talks

Banff Int'l Research Sta. *Workshop on Kernal Approximation and Gaussian Processes* 2026
CodEx Seminar *CodEx Seminar* 2026
UT Austin *Oden Institute Seminar* 2026
UT Austin *Data & Algebra Seminar* 2026
SIAM *Uncertainty Quantification* 2026
SIAM *Computational Science and Engineering (Minisymposium co-organizer)* 2025

Princeton University	*PACM IDeAS Seminar*	2025
SIAM	*Uncertainty Quantification*	2024
ICIAM	*International Congress on Industrial and Applied Mathematics*	2023
SIAM	*Mathematics of Data Science (Minisymposium co-organizer)*	2022

Teaching

Diff. Eq. with Linear Algebra	*UT Austin M427J*	Instructor	Fall 2026
Preparation for Calculus	*UT Austin M305G (TPEI)*	Instructor	Fall 2026
Differential Calculus	*UT Austin M408K (TPEI)*	Co-instructor	Spring 2026
Mathematic Statistics	*NYU MATH-UA.2340*	Teaching Assistant	Spring 2024
Statistics	*NYU MATH-GA.2962*	Grader	Fall 2021
Computational Statistics	*NYU MATH-GA.2080*	Grader	Spring 2021

Software

Primary developer	`FastHankelTransform.jl`, `SpectralKernels.jl`
Contributor	`FastGaussQuadrature.jl`, `chunkIE`

Outreach and Service

Peer reviewer
IEEE Signal Processing Letters, *Statistics and Computing*, *Journal of Machine Learning Research*, *SIAM Journal on Scientific Computing*.

Volunteer Instructor *Texas Prison Education Initiative (TPEI)* 2026-present
Instructor for credit-bearing college math courses for incarcerated youth and adults

Volunteer Instructor *Petey Greene Program* 2020-2025
Instructor for elementary through college level math for currently and formerly incarcerated adults

Reading Group Founder *Courant Institute* 2020-2024
Co-founded a department group reading articles on progress towards more accessible math higher education

Department Climate Survey Designer *Courant Institute* 2020-2022
Co-designed, administered, and reviewed a survey on Courant Institute PhD student experience